

Formative Assessment 4

```
suppressPackageStartupMessages({  
  library(readxl)  
  library(tidyverse)  
  library(car)  
  library(PerformanceAnalytics)  
  library(xts)  
  library(rugarch)  
  library(quantmod)  
  library(tseries)  
  library(forecast)  
  library(FinTS)  
})
```

Stock Returns

```
stock <- read_csv('dataset_stock_returns.csv', show_col_types = FALSE)  
stock$dates <- seq.Date(from = as.Date("2010-01-01"), by="days", length.out = nrow(stock))  
head(stock)
```

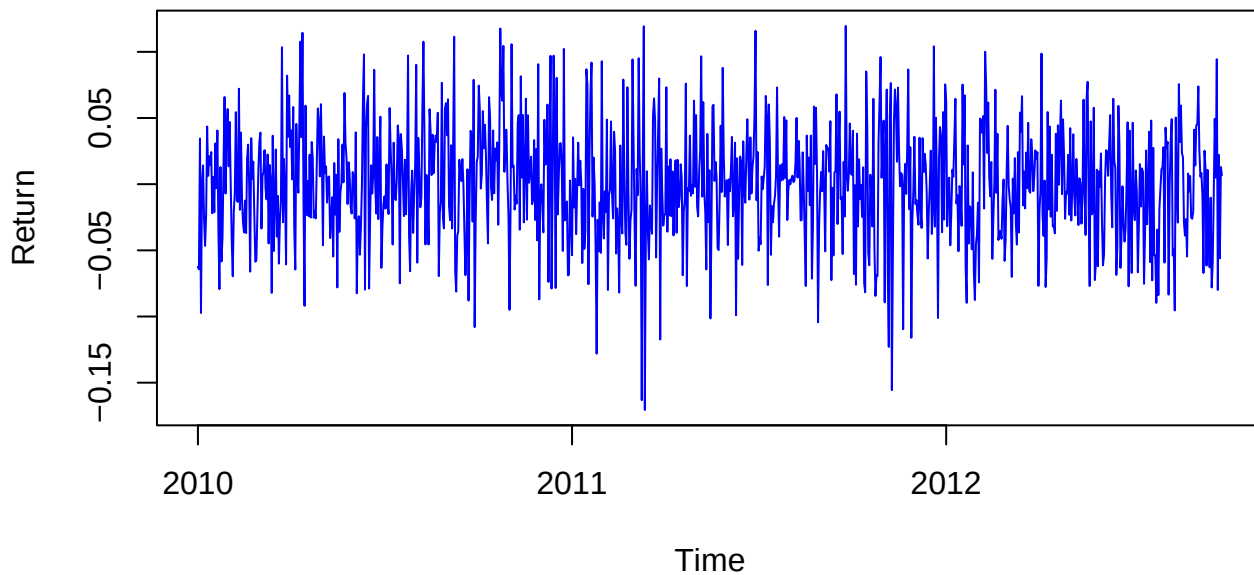
```
## # A tibble: 6 x 2  
##   returns dates  
##   <dbl> <date>  
## 1 -0.0625 2010-01-01  
## 2 -0.0643 2010-01-02  
## 3  0.0345 2010-01-03  
## 4 -0.0974 2010-01-04  
## 5  0.000708 2010-01-05  
## 6  0.0142 2010-01-06
```

Visual Exploration of Volatility

Return Series and Rolling Standard Deviation

```
plot(stock$dates, stock$returns, type = "l", col = "blue",  
      main = "Returns Series", xlab = "Time", ylab = "Return")
```

Returns Series



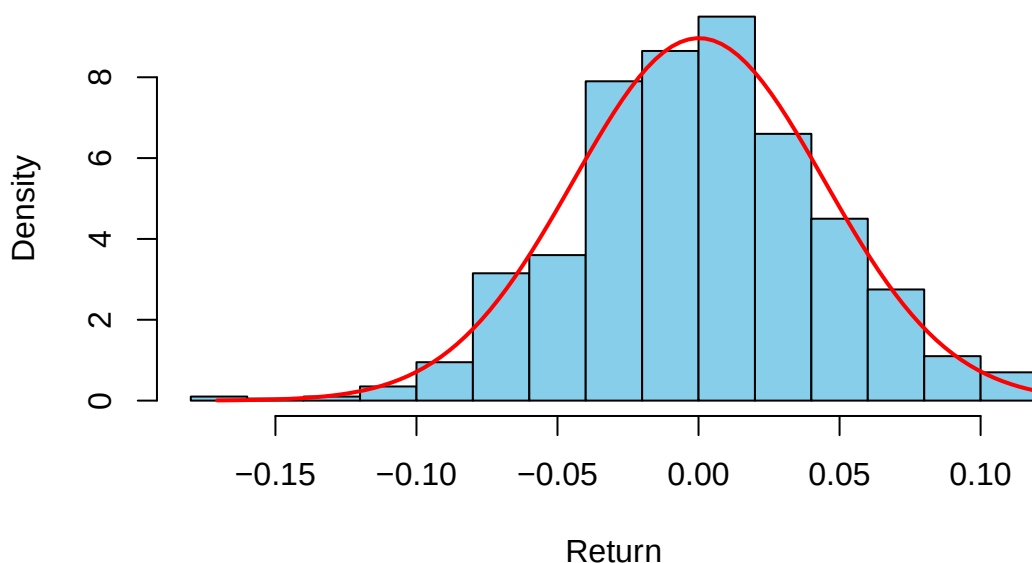
The return series plot shows returns fluctuating randomly around 0, with no apparent trend or seasonality and showing alternating periods of high and low variability. Notably, two distinct time intervals stand out where returns exhibit large negative and positive movements, preceded and followed by periods of relatively small changes. This pattern is indicative of volatility clustering.

```
hist(stock$returns, main="Distribution of Stock Returns",
     col = "skyblue", xlab= "Return", freq=FALSE)

x <- seq(min(stock$returns, na.rm = TRUE),
        max(stock$returns, na.rm = TRUE),
        length = 100)

lines(x, dnorm(x, mean = mean(stock$returns, na.rm = TRUE),
              sd = sd(stock$returns, na.rm = TRUE)), col="red", lwd=2)
```

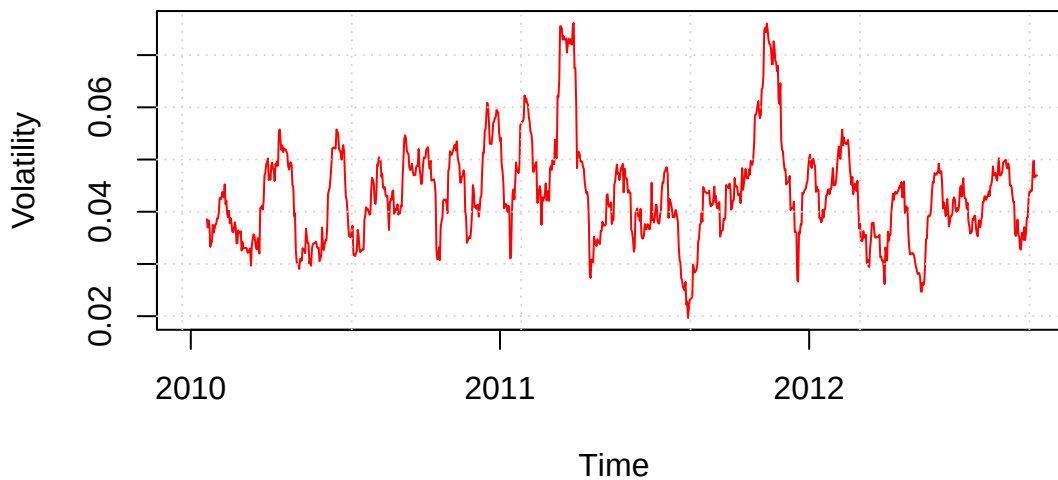
Distribution of Stock Returns



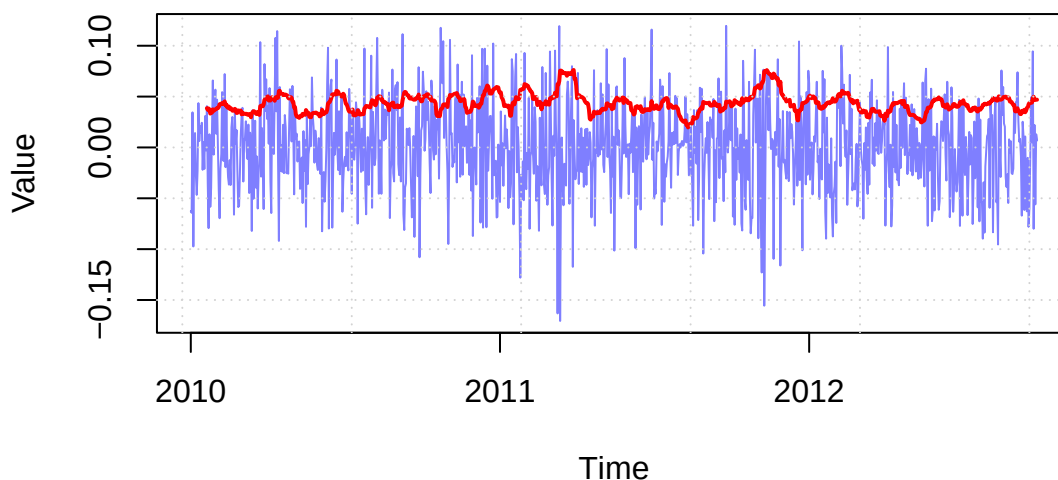
The plot of returns is not symmetrically distributed, rather it is negatively skewed. This indicates that large negative returns occur more frequently than large positive ones. Compared to the normal distribution, the returns have moderate kurtosis, meaning that extreme changes happen moderately.

```
par(mfrow=c(2,1))
rolling_sd <- runSD(stock$returns, n = 20, cumulative = FALSE)
plot(stock$dates, rolling_sd, type = "l", col = "red",
      main = "Rolling Standard Deviation (n=20)", xlab = "Time", ylab = "Volatility")
grid()
plot(stock$dates, stock$returns, type = "l", col = rgb(red = 0, green = 0, blue = 1, alpha = 0.5),
      main = "Returns and Rolling Standard Deviation (n=20)",
      xlab = "Time", ylab = "Value")
lines(stock$dates, rolling_sd, col = "red", lwd = 2)
grid()
```

Rolling Standard Deviation (n=20)



Returns and Rolling Standard Deviation (n=20)

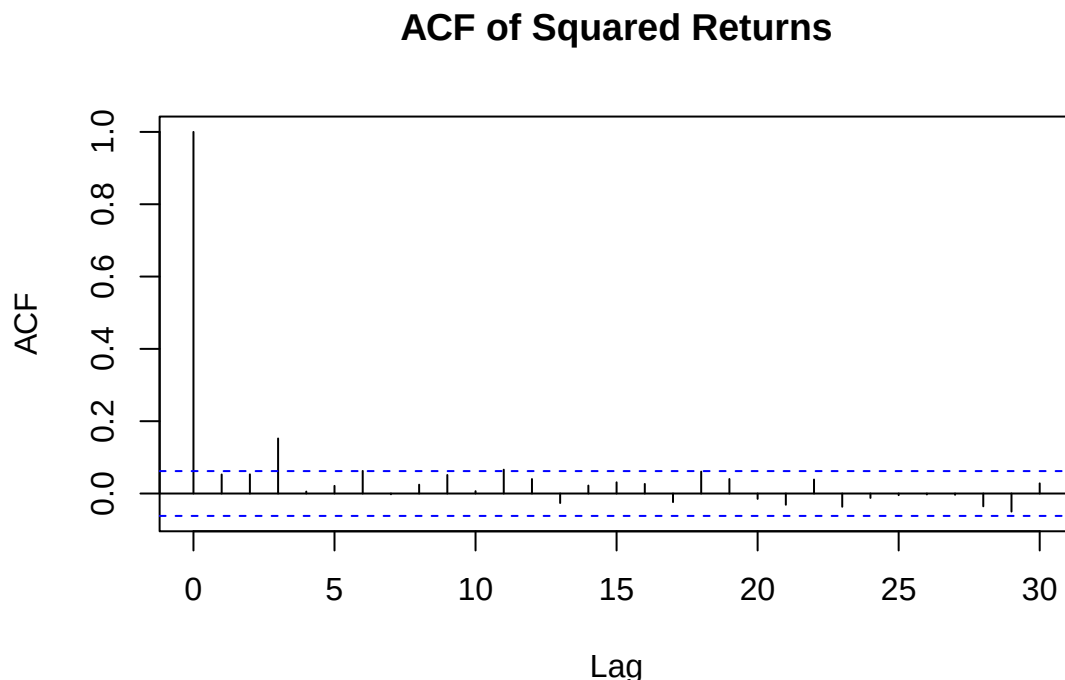


Assuming 20 trading days per month, the rolling standard deviation line shows the monthly volatility of the stock returns. It rises sharply during turbulent periods, reflecting increased volatility, and falls during calm periods when

the market stabilizes. This behavior suggests that the variance of returns is not constant over time, implying conditional heteroskedasticity.

Plot ACF of Squared Returns

```
acf((stock$returns)^2, main = "ACF of Squared Returns")
```



The ACF of squared returns drops quickly to zero and is insignificant at lags 1–2 but becomes significant again at lag 3. This pattern suggests an absence of immediate volatility clustering, but the presence of some delayed or periodic dependence in volatility roughly every three periods. The changing variance of returns over time indicates conditional heteroskedasticity, consistent with the presence of volatility dynamics. This could point to a potential ARCH(3)-type effect, where volatility depends on shocks occurring three periods earlier.

```
ArchTest(stock$returns, lags = 12)
```

```
##
##  ARCH LM-test; Null hypothesis: no ARCH effects
##
## data:  stock$returns
## Chi-squared = 33.849, df = 12, p-value = 0.0007124
```

Model Estimation

Model 1: ARCH(3)

The ACF of squared returns shows a small but noticeable dependence up to lag 3, suggesting that volatility today may be influenced by shocks over the past three periods. The ARCH(3) model directly captures this short-term volatility dependence by including three lag terms.

```
arch3_spec <- ugarchspec(
  variance.model = list(model = "sGARCH", garchOrder = c(3,0)),
  mean.model      = list(armaOrder = c(0,0), include.mean = TRUE),
  distribution.model = "std")
```

```
arch3_fit <- ugarchfit(spec = arch3_spec, data = stock)
```

Coefficients and Standard Errors

```
arch3_fit@fit$matcoef
```

##		Estimate	Std. Error	t value	Pr(> t)
## mu		-4.345627e-05	1.380715e-03	-0.03147375	0.974891726
## omega		1.633322e-03	1.294728e-04	12.61517245	0.000000000
## alpha1		2.153733e-02	4.099328e-02	0.52538675	0.599314360
## alpha2		4.071563e-02	3.020692e-02	1.34789059	0.177693575
## alpha3		1.073469e-01	4.130901e-02	2.59863226	0.009359598
## shape		9.454410e+01	1.539230e+02	0.61422997	0.539063373

The mean μ term is insignificant, suggesting that daily stock returns fluctuate randomly around zero, reflecting the uncertainty in market. The constant term ω is small but highly significant ($p < 0.001$), indicating a low baseline level of market volatility during stable periods. Among the three ARCH coefficients, only the third lag ($\alpha_3 = 0.107, p = 0.009$) is statistically significant. This suggests that volatility is primarily influenced by shocks from three days prior, indicating a delayed reaction to past shocks rather than an immediate response. The shape parameter is also insignificant, suggesting that the residuals do not strongly deviate from normality. These results imply that market participants remain cautious or risk-averse for some days following major movements.

Model 2: GARCH(1,1)

This model extends the original ARCH framework by allowing current volatility to depend on recent shocks (ARCH term) and recent volatility (GARCH term).

```
garch11_spec <- ugarchspec(
  variance.model = list(model = "sGARCH", garchOrder = c(1, 1)),
  mean.model = list(armaOrder = c(0, 0), include.mean = TRUE),
  distribution.model = "std"
)

garch11_fit <- ugarchfit(spec = garch11_spec, data = stock)
```

Coefficient Estimates and Standard Errors

```
garch11_fit@fit$matcoef
```

##		Estimate	Std. Error	t value	Pr(> t)
## mu		1.624450e-04	1.374498e-03	0.1181849	0.90592112
## omega		2.146653e-04	1.128429e-04	1.9023380	0.05712699
## alpha1		5.414629e-02	2.157305e-02	2.5099038	0.01207641
## beta1		8.369426e-01	6.914530e-02	12.1041147	0.00000000
## shape		7.557222e+01	1.236757e+02	0.6110513	0.54116562

Again, the μ term is insignificant in this model, while the constant ω term is small and is only marginally significant ($p < 0.1$), implying a low but nonzero long-run variance that forms the baseline level of market volatility. The ARCH coefficient ($\alpha_1 = 0.054, p = 0.01$) is positive and significant, suggesting that recent shocks have a considerable effect on current volatility. Comparably, the GARCH coefficient ($\beta_1 = 0.837, p < 0.001$) is relatively large and significant, indicating that past volatility strongly persists into the current period. Together, the ARCH and GARCH terms imply that volatility is highly persistent with shocks dissipating after some days. This reflects investors' cautious behavior, delayed market reactions, and ongoing uncertainty after large movements.

AIC Comparison

```
data.frame(  
  Model = c("ARCH(3)", "GARCH(1,1)"),  
  AIC = c(Infocriteria(arch3_fit)[1], Infocriteria(garch11_fit)[1]),  
  BIC = c(Infocriteria(arch3_fit)[2], Infocriteria(garch11_fit)[2])  
)
```

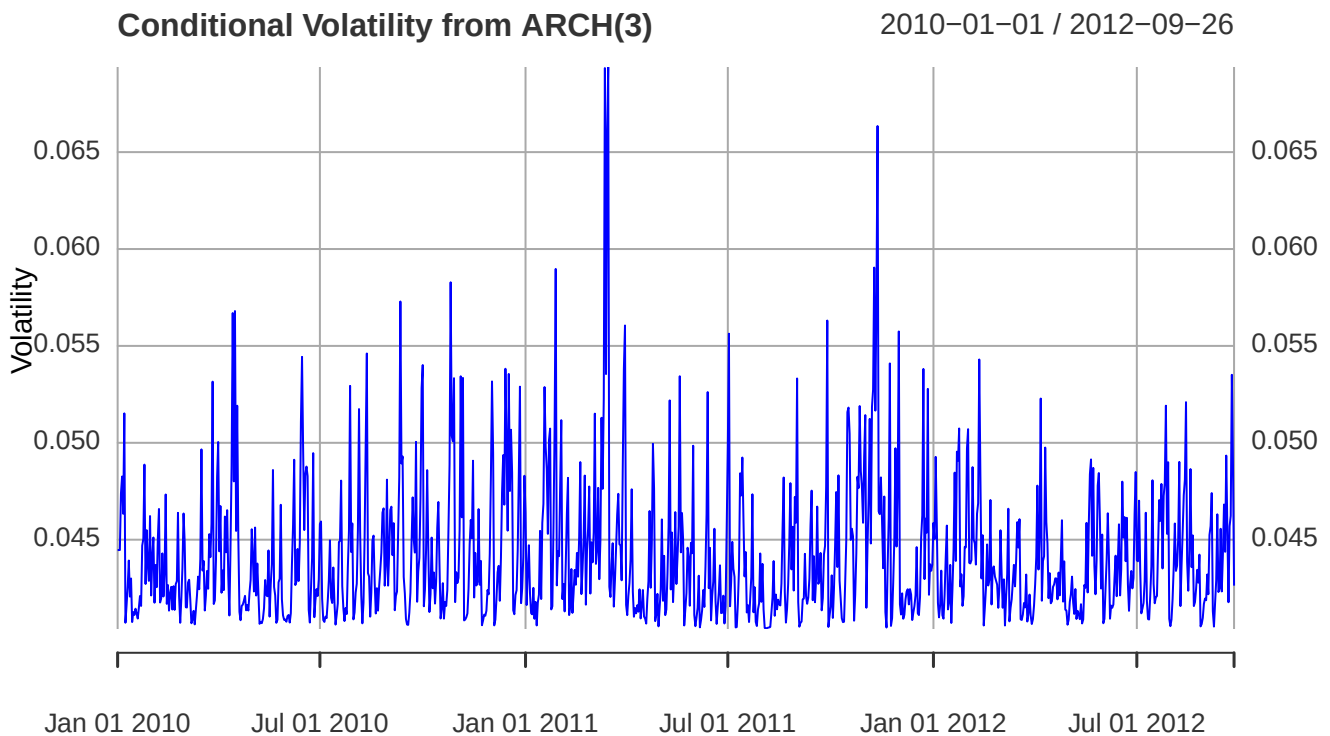
```
##      Model      AIC      BIC  
## 1   ARCH(3) -3.392429 -3.362983  
## 2 GARCH(1,1) -3.393307 -3.368768
```

While both models have very similar AIC and BIC values, GARCH(1,1) has slightly lower values, indicating a marginally better fit. Furthermore, GARCH(1,1) is able to capture both short-term shock effects and persistent volatility, aligning with the observed volatility clustering in the returns series and volatility plots. Meanwhile, ARCH(3) only captures shocks up to three days ago and misses the longer-term persistence, which is why its fit is slightly worse.

Diagnostics and Volatility Interpretation

Conditional Volatility

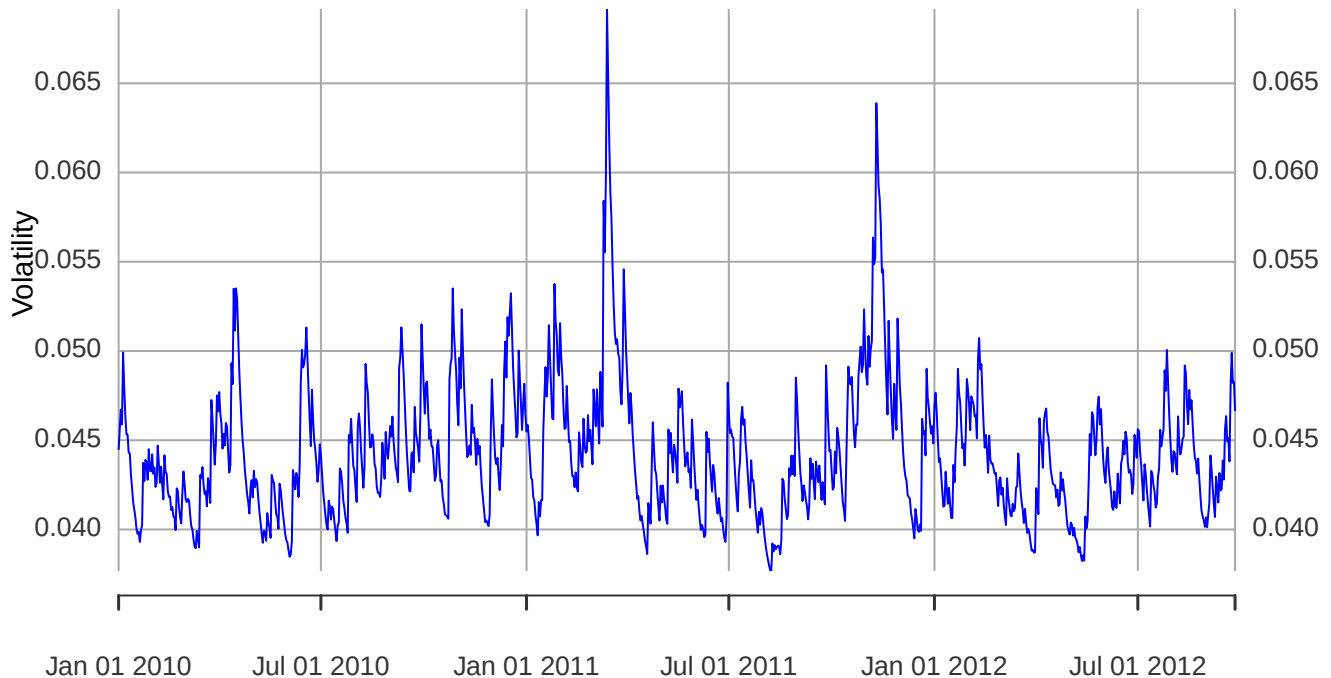
```
plot(sigma(arch3_fit), main= "Conditional Volatility from ARCH(3)",  
      ylab="Volatility", col="blue", lwd=1)  
grid()
```



```
plot(sigma(garch11_fit), main= "Conditional Volatility from GARCH(1,1)",  
      ylab="Volatility", col="blue", lwd=1)  
grid()
```

Conditional Volatility from GARCH(1,1)

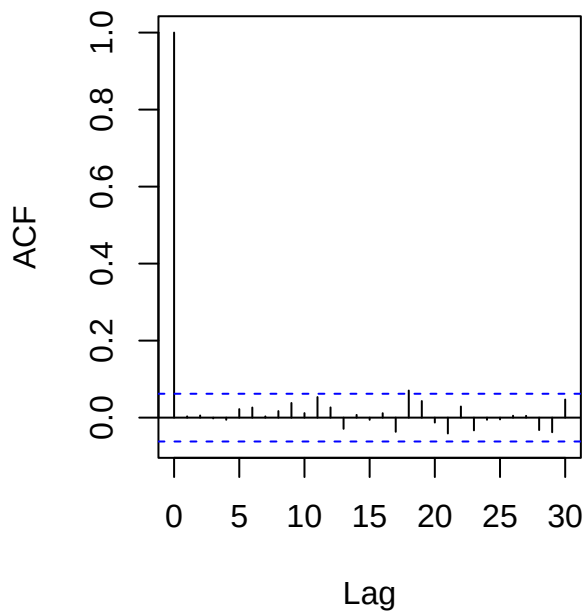
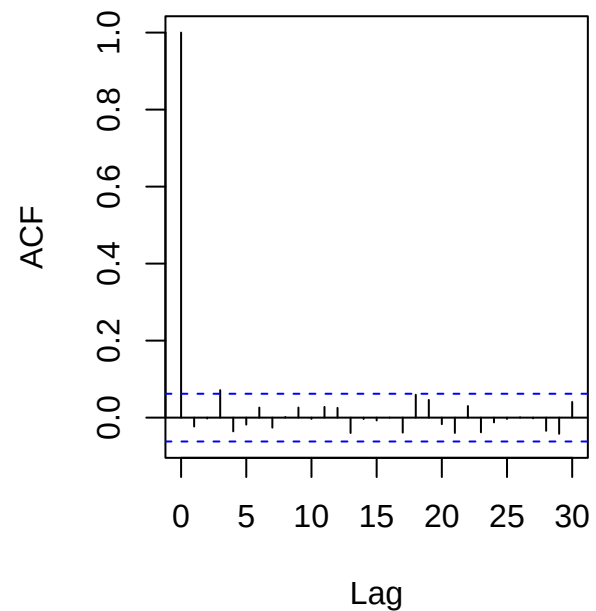
2010-01-01 / 2012-09-26



As ARCH(3) only relies on past shocks, it produced a more jagged volatility series. It responds more sharply to individual shocks and drops back to the baseline quickly, which may underestimate volatility. In contrast, the GARCH(1,1) provides a smoother visualization of volatility decaying realistically, demonstrating the persistence of volatility.

ACF of Squared Residuals

```
arch3_resid <-residuals(arch3_fit, standardize=TRUE)
garch11_resid <-residuals(garch11_fit, standardize=TRUE)
par(mfrow=c(1,2), mai = c(1, 0.8, 0.7, 0.5))
acf(arch3_resid^2, main ="ACF of ARCH(3) Residuals^2")
acf(garch11_resid^2, main ="ACF of GARCH(1,1) Residuals^2")
```

ACF of ARCH(3) Residuals²**ACF of GARCH(1,1) Residuals²**

ARCH-LM test

```
ArchTest(arch3_resid)
```

```
##
##  ARCH LM-test; Null hypothesis: no ARCH effects
##
## data:  arch3_resid
## Chi-squared = 6.7278, df = 12, p-value = 0.8751
```

```
ArchTest(garch11_resid)
```

```
##
##  ARCH LM-test; Null hypothesis: no ARCH effects
##
## data:  garch11_resid
## Chi-squared = 9.4164, df = 12, p-value = 0.667
```

The ACF plots show no remaining ARCH effect in the residuals, indicating a good fit for both models. The ARCH-LM test was also performed to formally assess the presence of ARCH effects in the residuals. Results for both models yielded p-values above the standard significance threshold of 0.05, confirming the findings from the ACF plots.

Because stock returns typically exhibit long-lasting volatility clustering, the GARCH(1,1) model offers a more realistic representation of the underlying volatility dynamics over the simple ARCH model.

Volatility Forecasting

Forecast conditional volatility for 10 future periods


```

arch3_fc <- ugarchforecast(fitORspec = arch3_fit, n.ahead=10)
garch11_fc <- ugarchforecast(fitORspec =garch11_fit, n.ahead=10)

arch3_fc_vol <- sigma(arch3_fc)
garch11_fc_vol <- sigma(garch11_fc)

fc <- data.frame(arch3_fc_vol,garch11_fc_vol)
colnames(fc) <- c("ARCH(3)", "GARCH(1,1)")
fc

```

```

##          ARCH(3) GARCH(1,1)
## T+1  0.04446677 0.04513240
## T+2  0.04118233 0.04505280
## T+3  0.04189766 0.04498174
## T+4  0.04418640 0.04491833
## T+5  0.04391928 0.04486175
## T+6  0.04407718 0.04481127
## T+7  0.04430902 0.04476624
## T+8  0.04429187 0.04472608
## T+9  0.04431775 0.04469026
## T+10 0.04434242 0.04465832

```

Forecasted volatility path with confidence intervals

```

plot_fc <- function(fit, fc) {
  hist_vol <- tail(sigma(fit), 30)
  fc_vol <- sigma(fc)
  fc_mean <- fitted(fc)
  hist_dates <- index(hist_vol)
  fc_dates <- seq(from = hist_dates[length(hist_dates)] + 1,
                  by = "days", length.out = 10)

  upper <- fc_vol * 1.96
  lower <- fc_vol * -1.96

  all_vol <- c(as.numeric(hist_vol), fc_vol)
  all_dates <- c(hist_dates, fc_dates)

  plot(all_dates, all_vol, type = "n", col = "blue", lwd = 2,
       ylim = range(c(lower, upper, fc_vol)),
       main = "10-Day Forecasted Conditional Volatility",
       xlab = "Time", ylab = "Volatility")

  polygon(c(fc_dates, rev(fc_dates)),
         c(upper, rev(lower)),
         col = "lightblue", border = NA)

  lines(hist_dates, hist_vol, col = "black")
  lines(fc_dates, fc_vol, col = "blue", lty = 2)

  legend("bottomleft", legend = c("Last 30 Days", "Forecast"),
        col = c("black", "blue"), lwd = c(1, 1), lty = c(1, 2))
}

```

```

plot_fc <- function(fit, fc) {
  hist_vol <- tail(sigma(fit), 90)
  hist_dates <- index(hist_vol)

  fc_var <- fc@forecast$sigmaFor^2
  fc_se <- fc@forecast$varFor

  if (is.null(fc_se) || all(is.na(fc_se))) {
    var_upper <- fc_var * 1.2
    var_lower <- fc_var * 0.8
  } else {
    var_upper <- pmax(fc_var + 1.96 * fc_se, 0)
    var_lower <- pmax(fc_var - 1.96 * fc_se, 0)
  }

  vol_fc <- sqrt(fc_var)
  vol_upper <- sqrt(var_upper)
  vol_lower <- sqrt(var_lower)

  fc_dates <- tail(hist_dates, 1) + 1:length(vol_fc)

  all_dates <- c(hist_dates, fc_dates)
  all_vol <- c(as.numeric(hist_vol), vol_fc)

  plot(all_dates, all_vol, type = "n",
       ylim = range(c(vol_lower, vol_upper), na.rm = TRUE),
       main = "Forecasted Conditional Volatility",
       xlab = "Time", ylab = "Volatility")

  polygon(c(fc_dates, rev(fc_dates)),
         c(vol_upper, rev(vol_lower)),
         col = "lightblue", border = NA)

  lines(hist_dates, hist_vol, col = "black")
  lines(fc_dates, vol_fc, col = "blue", lty = 2)

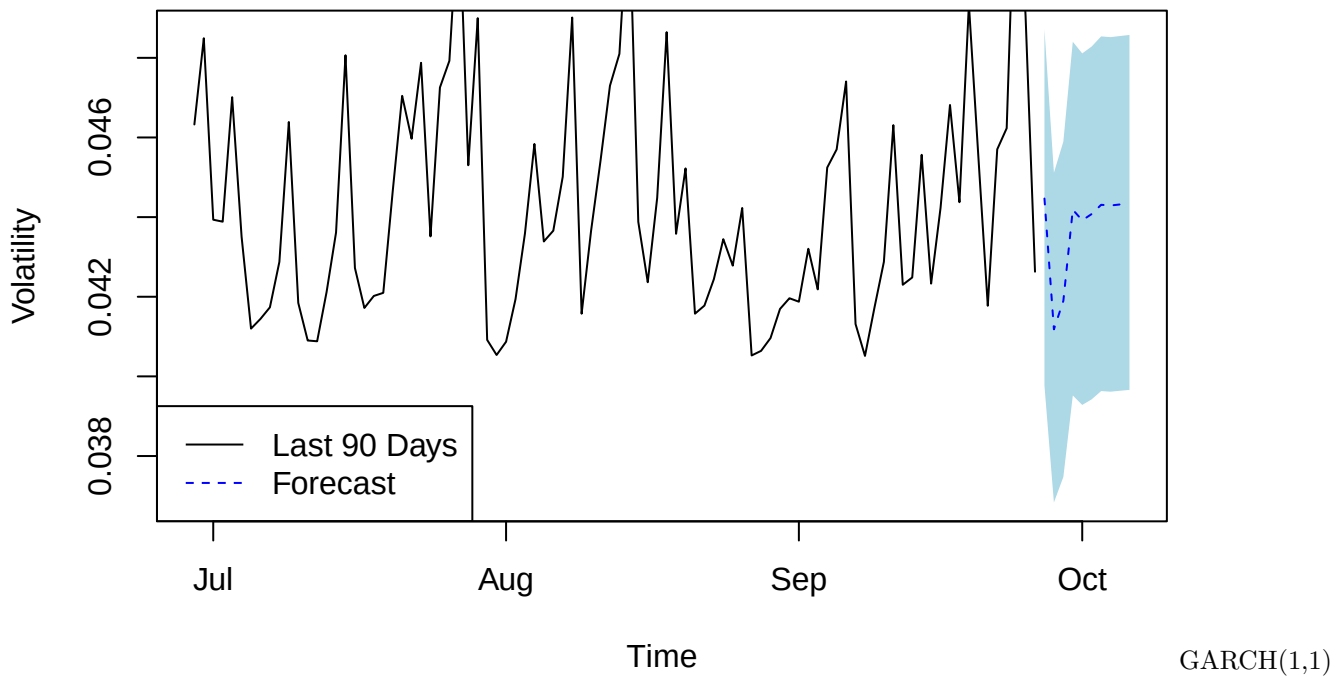
  legend("bottomleft", legend = c("Last 90 Days", "Forecast"),
       col = c("black", "blue"), lwd = c(1,1), lty = c(1,2))
}

```

ARCH(3)

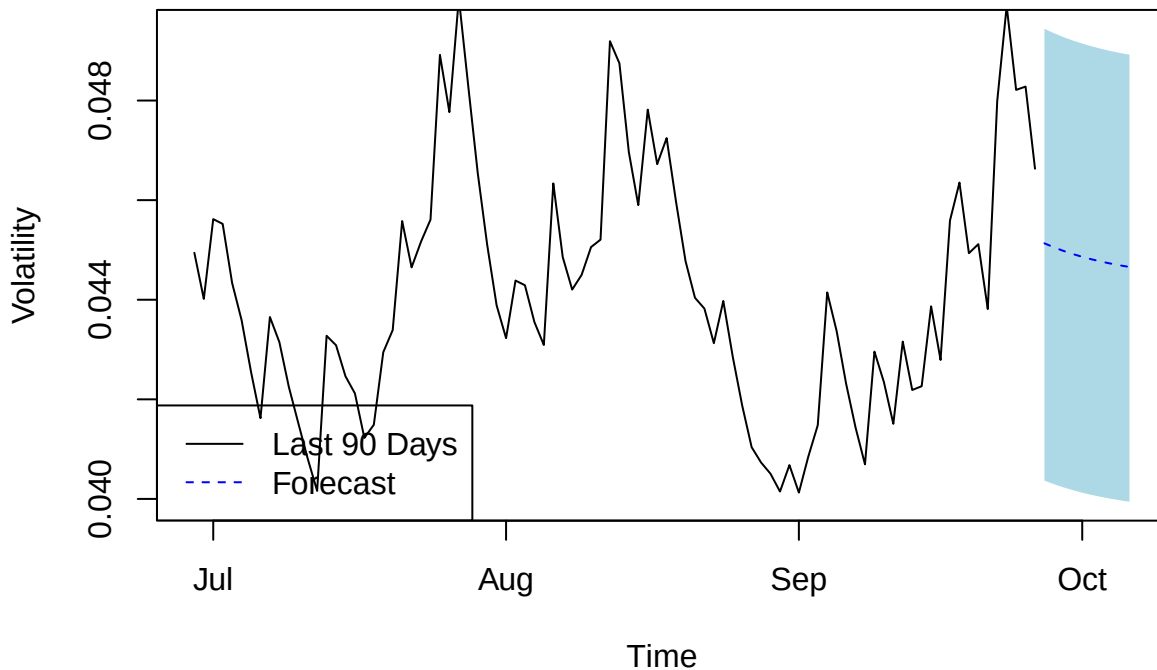
```
plot_fc(arch3_fit, arch3_fc)
```

Forecasted Conditional Volatility



```
plot_fc(garch11_fit, garch11_fc)
```

Forecasted Conditional Volatility



The plot shows the conditional volatility over the last three months (90 days) alongside the 10-day forecasted volatility from both the ARCH(3) and GARCH(1,1) models. As expected, the forecasted volatility path generated by ARCH(3) shows the model's sensitivity to past shocks, resulting in quick fluctuations. In contrast, GARCH(1,1) model's forecast exhibits a smoother decay, reflecting the model's ability to capture volatility persistence. Both forecasts gradually decay over the 10-day horizon, reflecting the natural tendency of conditional volatility to revert toward its long-term average after recent shocks.

The wide forecast intervals for both models, spanning roughly same width as the range of the historical data, leave much room for uncertainty in future volatility estimates.

Reflection and Application

Volatility modeling is highly relevant in finance, where stock returns exhibit time-varying uncertainty that affects pricing, risk management, and portfolio optimization. Historical stock returns can be analyzed to capture patterns like volatility clustering (periods of high or low volatility) by relating current volatility to past returns and past volatility. ARCH models can capture short bursts of volatility but may struggle in showing the persisting effects of events. GARCH improves on this by being able to capture both short and long-run volatility persistence. Both models, however, may be less effective when returns exhibit asymmetry, which was also present in the recorded data. For further improvement on this, EGARCH models may be explored for its ability to capture asymmetric responses to shocks.

Bitcoin Returns

```
bitcoin <- read_csv('dataset_bitcoin_returns.csv', show_col_types = FALSE)
bitcoin$dates <- seq.Date(from = as.Date("2010-01-01"), by="days", length.out = nrow(bitcoin))

head(bitcoin)
```

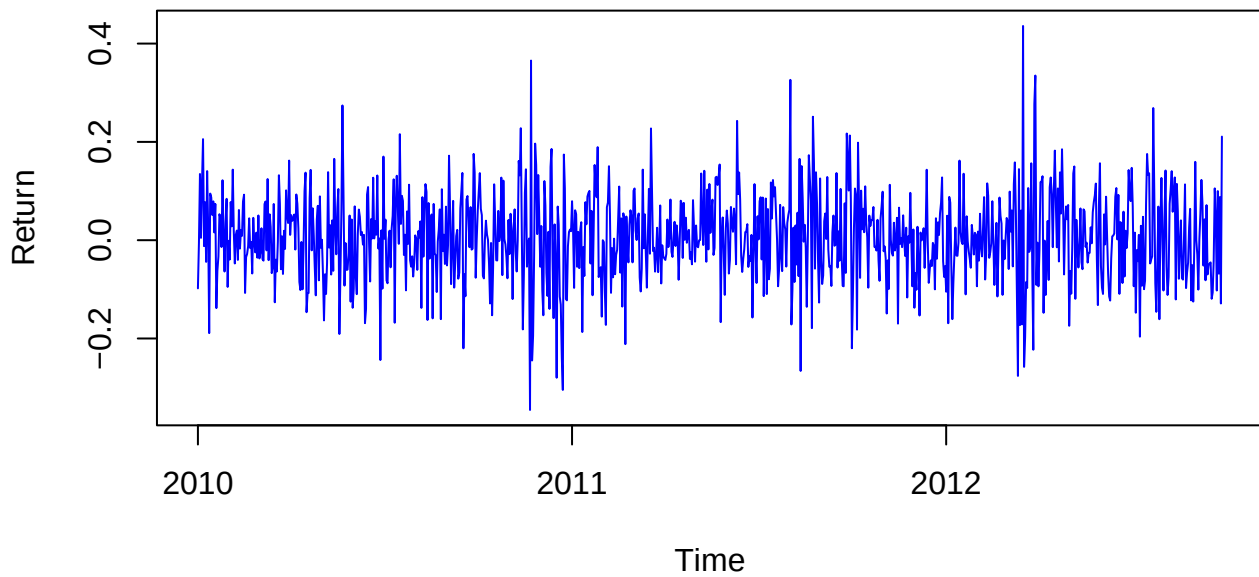
```
## # A tibble: 6 x 2
##   returns dates
##   <dbl> <date>
## 1 -0.0980 2010-01-01
## 2 -0.00617 2010-01-02
## 3  0.135   2010-01-03
## 4  0.00469 2010-01-04
## 5  0.119   2010-01-05
## 6  0.206   2010-01-06
```

Visual Exploration of Volatility

Return Series and Rolling Standard Deviation

```
plot(bitcoin$dates, bitcoin$returns, type = "l", col = "blue",
     main = "Bitcoin Returns Series", xlab = "Time", ylab = "Return")
```

Bitcoin Returns Series

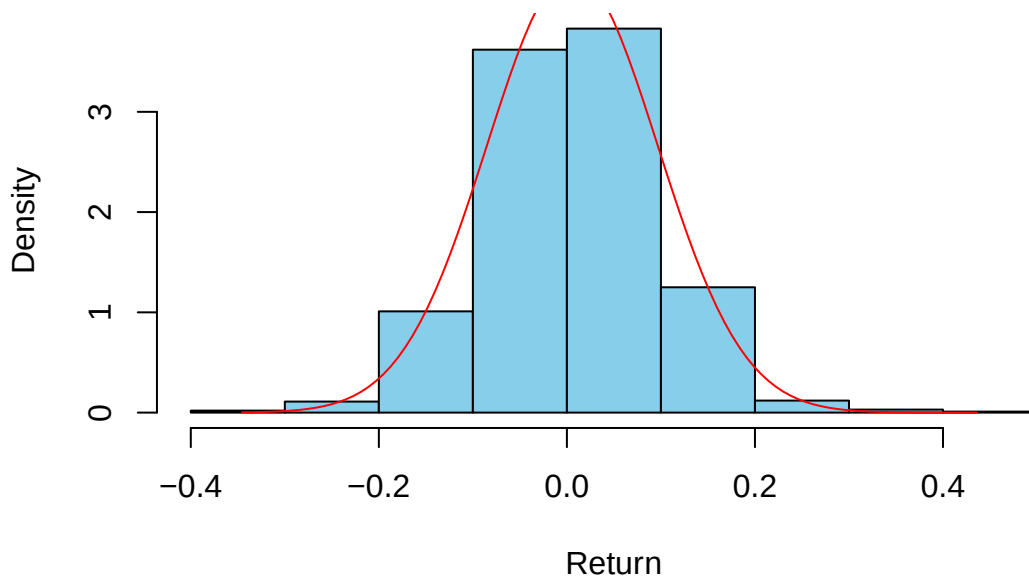


```
hist(bitcoin$returns, main="Distribution of Bitcoin Returns",
     col = "skyblue", xlab= "Return", freq = FALSE)

x <- seq(min(bitcoin$returns, na.rm = TRUE),
        max(bitcoin$returns, na.rm = TRUE),
        length = 100)

lines(x, dnorm(x, mean = mean(bitcoin$returns, na.rm = TRUE),
              sd = sd(bitcoin$returns, na.rm = TRUE)), col="red")
```

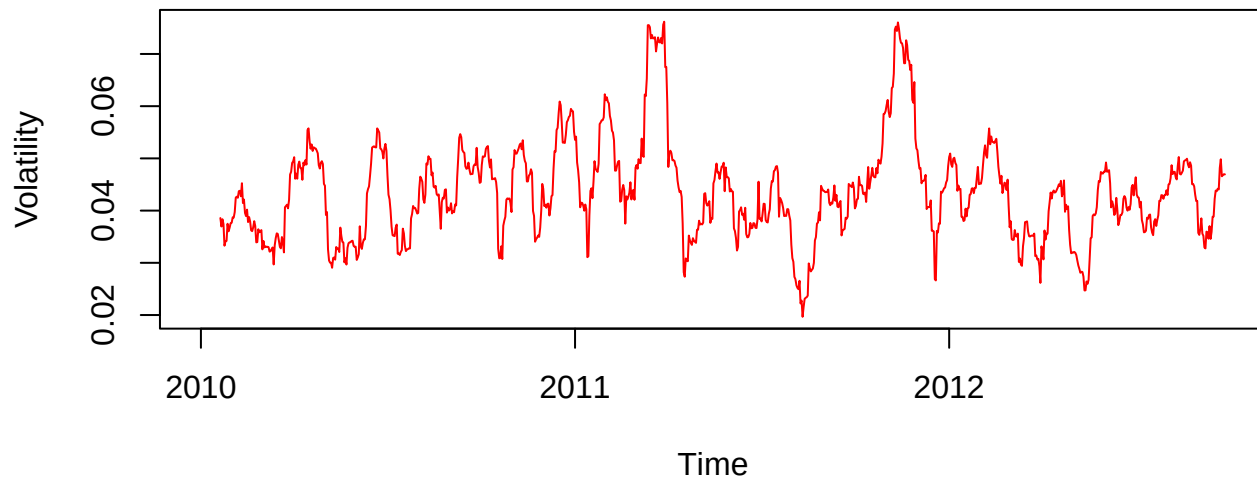
Distribution of Bitcoin Returns



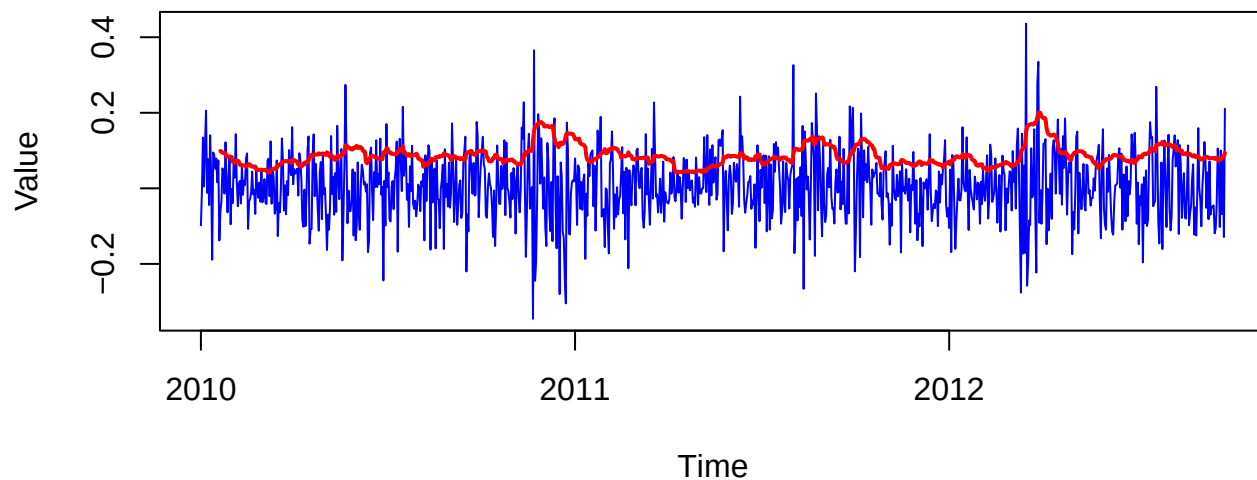
```
par(mfrow=c(2,1))
plot(bitcoin$dates, rolling_sd, type = "l", col = "red",
     main = "Rolling Standard Deviation (n=20)", xlab = "Time", ylab = "Volatility")
```

```
rolling_sd <- runSD(bitcoin$returns, n = 20, cumulative = FALSE)
plot(bitcoin$dates, bitcoin$returns, type = "l", col = "blue",
     main = "Returns and Rolling Standard Deviation (n=20)",
     xlab = "Time", ylab = "Value")
lines(bitcoin$dates, rolling_sd, col = "red", lwd = 2)
```

Rolling Standard Deviation (n=20)

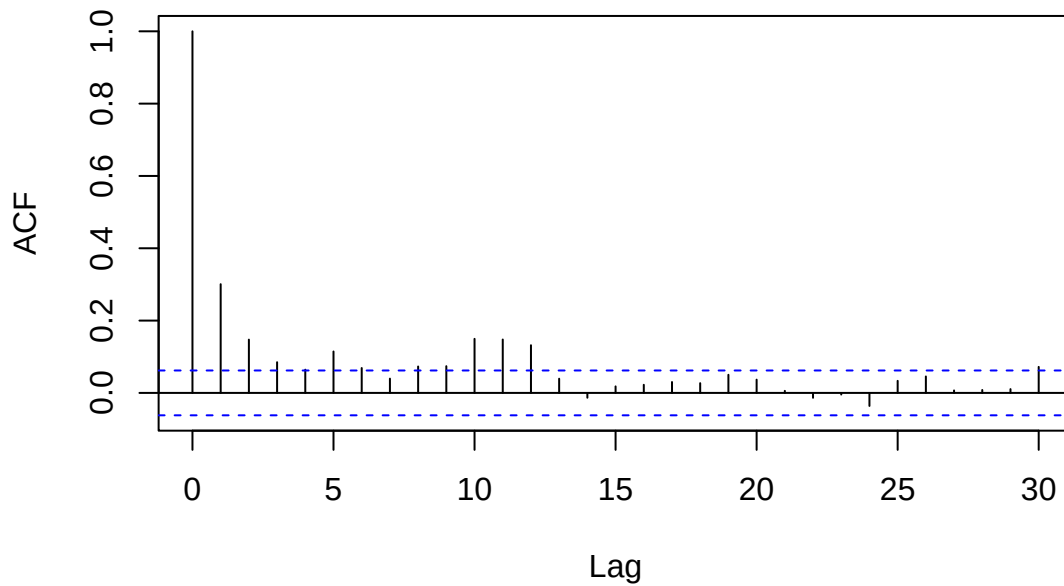


Returns and Rolling Standard Deviation (n=20)



```
acf((bitcoin$returns)^2, main = "ACF of Squared Returns")
```

ACF of Squared Returns



Slow decay, pattern

```
ArchTest(bitcoin$returns, lags = 12)
```

```
##  
## ARCH LM-test; Null hypothesis: no ARCH effects  
##  
## data: bitcoin$returns  
## Chi-squared = 127.73, df = 12, p-value < 2.2e-16
```

Model Estimation

Model 1 GARCH(1,1)

```
garch11_fit <- ugarchfit(garch11_spec, data = bitcoin$returns)
```

```
garch11_fit@fit$matcoef
```

```
##           Estimate Std. Error  t value    Pr(>|t|)  
## mu      4.801024e-03 0.002603520  1.844051 0.0651756744  
## omega   5.481553e-04 0.000254411  2.154605 0.0311927474  
## alpha1  9.847852e-02 0.026037228  3.782220 0.0001554359  
## beta1   8.349503e-01 0.048847206 17.093102 0.0000000000  
## shape   1.812666e+01 8.248175087  2.197657 0.0279735591
```

```
infocriteria(garch11_fit)
```

```
##  
## Akaike      -2.030367  
## Bayes      -2.005828  
## Shibata    -2.030416  
## Hannan-Quinn -2.021040
```

Model 2

Record estimated parameters, standard errors, and information criteria (AIC, BIC).

Diagnostics and Volatility Interpretation

Plot conditional volatility and compare across models.

Analyze residuals for remaining ARCH effects (using ACF of residuals squared or ARCH-LM test).

Comment on which model better captures volatility persistence and asymmetry.

Volatility Forecasting

Forecast conditional volatility for 10 future periods

forecasted volatility path with confidence intervals

what happens to volatility over the forecast horizon (does it decay, persist, or show asymmetry?)

Reflection and Application

Discuss in which contexts (finance, energy, currency) volatility modeling is most relevant.

Identify any limitations or possible model improvements.